

Internal and External Focus of Attention in a Novice Form Sport

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In the current experiment, we examined optimal focus for novices during a movement sequence in which performance was measured on accurate movement form/technique. A novel gymnastics routine was practiced under either an internal skill-relevant, internal skill-irrelevant, external, or no attention focus. Retention and transfer tests were then completed. During acquisition, adopting an internal irrelevant focus significantly improved performance, whereas an external focus degraded performance. There were no significant group differences in the retention and transfer tests. This suggests that learning of movement form/technique did not benefit from a specific focus of attention. The results are interpreted via an attentional capacity viewpoint and the notion that form tasks do not always contain obvious movement effects central to common coding and the constrained action hypothesis.

Key words: gymnastics, skill acquisition, technique, verbal instruction

A large body of research concerns how a performer's focus of attention influences motor learning and control (for reviews, see Wulf & Prinz, 2001; Wulf, 2007a). The results of much of this research concur with early suggestions by James (1890), who stated "keep your eye at the place aimed at, and your hand will fetch the target; think of your hand, and you will likely miss your aim" (p. 520). Here James proposed that adopting a focus on the body will negatively affect performance, whereas adopting a focus on the effects one's movements have on the environment (i.e., the place aimed at, or target) will enhance performance. In more recent research, these different foci of attention have been defined as internal (focusing on particular body parts while executing a skill) and external (focusing on a movement effect; Wulf, McNevin, Fuchs, Ritter, & Toole, 2000).

The benefit of adopting an external focus of attention has been demonstrated in a variety of motor skills,

including basketball (Al-Abood, Bennett, Hernandez, Ashford, & Davids, 2002; Zachry, Wulf, Mercer, & Bezodis, 2005), jumping (Wulf, Zachry, Granados, & Dufek, 2007), golf (Wulf, Lauterbach, & Toole, 1999), balance (Wulf, McNevin, & Shea, 2001), and leg cycling (Totsika & Wulf, 2003). Furthermore, this benefit has been observed for both novice and expert performers (Bell & Hardy, 2009; Wulf & Su, 2007) and for individuals with movement disorders, such as stroke patients (Fasoli, Trombly, Tickle-Degnen, & Verfaellie, 2002) and Parkinson's disease patients (Landers, Wulf, Wallmann, & Guadagnoli, 2005). In past research, performance was measured via an outcome criterion (e.g., speed and accuracy). In the present experiment, we examined the performance of novices in a gymnastics routine under different foci of attention to examine whether the previous findings can be generalized to form-based tasks where technique rather than outcome is the primary determinant of performance.

According to the action-effect principle (Prinz, 1997), there is a compatible relationship between planning and outcome, with "actions planned and controlled in terms of their effects" (p. 152). Thus, if movements are planned in relation to their outcome then focusing on movement effects should augment performance, by enhancing efficiency of motor programming (Wulf, Höß & Prinz, 1998). Wulf et al. (2001) combined the notion of the action-effect principle with research involving focus of attention and proposed the constrained action hypothesis (CAH). Con-

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sistent with Prinz's (1997) action-effect principle, the CAH maintains that if performers focus on their movements, the congruence between movement programming and response diminishes. As a result, attending to movements of the body disrupts the organization of motor programming and interferes with normally automatic control processes. In contrast, adopting an external focus of attention enhances the efficiency of motor programming (by both increasing the relationship between movement planning and movement outcome and by preventing participants from consciously processing movements) and promotes automatic processing, leading to improved performance.

By assessing the attention demands of adopting either an internal or an external focus during movement execution, Wulf et al. (2001) investigated the proposal that an external focus of attention prevents participants from consciously processing movements. To this end, participants performed a balance task simultaneously with a probe reaction time task. Results at retention revealed that the external focus group displayed lower attention demands and superior balance performance compared to the internal focus group, thus supporting the CAH with respect to the predicted lower cognitive demands associated with external focus.

While Wulf and colleagues (Wulf et al., 1998; Wulf et al., 1999; Wulf et al., 2001; Wulf et al., 2007) revealed performance benefits through the adoption of an external focus of attention, numerous researchers have been unsuccessful when attempting to replicate these findings (Beilock, Carr, MacMahon, & Starkes, 2002; Ford, Hodges, & Williams, 2005; Perkins-Ceccato, Passmore, & Lee, 2003). These equivocal results have primarily been observed in research investigating novice and expert performers (Beilock et al., 2002; Ford et al., 2005; Wulf & Su, 2007). For example, Beilock et al. required right-footed expert and novice participants to perform a football (soccer) dribbling task under differing foci of attention: self-focus, where participants focused on their movements; and dual task, where participants were distracted from focusing on their movements to prevent explicit monitoring of the skill. Whereas novice performance was superior in the self-focus condition, optimal focus for expert performance depended on the use of either the dominant or nondominant foot. When using their dominant foot, experts' performance was superior under the dual task (distraction) condition. However, when using their nondominant foot, experts' performance was greatest under the self-focus condition.

Similar to Beilock et al. (2002), Ford et al. (2005) investigated expert and novice performance under different foci of attention. Participants performed a similar soccer dribbling task under one of four focus conditions: (a) internal skill-relevant focus, in which they focused on the foot they were dribbling with; (b) internal skill-irrelevant focus, in which their focus was on the arm; (c)

dual-task condition, in which participants were distracted from focusing on their movements by having to respond to words presented audibly; and (d) a control condition. The results for experts dribbling with their dominant foot were consistent with the CAH, as participants completed the dribbling task slower during both internal focus conditions compared to the dual-task condition. However, when experts were required to dribble with their nondominant foot, this effect was reversed, with movement times being slowest during the dual-task condition. For novices, dribbling times were slower during both the dual-task (distraction) and internal irrelevant (arm) conditions compared to the internal relevant condition (foot).

The findings of both Beilock et al. (2002) and Ford et al. (2005) have been explained by the deautomization-of-skills hypothesis (Beilock et al., 2002). Here it is reasoned that novice performance, since it is not yet automatic, cannot be disrupted through conscious control and is facilitated by allowing performers to attend to step-by-step processes of the skill. On the other hand, when experts adopt a self-focus, deautomization occurs through "reinvesting actions and percepts with attention" (Deikman, 1966, p. 329), resulting in decreased performance. Thus, when performance is proceduralized or automatic (i.e., expert), adopting an external focus leads to enhanced performance. On the other hand, when movements are less proceduralized, as in novices and experts using a less-favored limb, performance is superior when focus is directed internally to task relevant movements.

More recently, Wulf and Su (2007) re-examined the differential effects of internal and external focus of attention on expert and novice performance in a golf-putting task. The results revealed that both experts and novices performed better when directed to adopt an external focus of attention, as opposed to an internal focus of attention. These findings are inconsistent with the deautomization-of-skills hypothesis and the previous results of Beilock et al. (2002) and Ford et al. (2005). However, it should be noted that the external focus conditions adopted by Wulf and colleagues differed from those of Beilock et al. (2002) and Ford et al. (2005), with the former using direct verbal instructions (i.e., "focus on...") to manipulate external focus and the latter using an indirect manipulation of attention through the use of a distracting task (i.e., word-monitoring dual-task conditions). As such, the null effects of adopting an external focus of attention in the research of Beilock et al. (2002) and Ford et al. (2005) may have been due to the method by which focus was manipulated. For example, focusing attention externally through the use of direct verbal instructions could be considered more consistent with Prinz's (1997) initial action-effect principle compared to adopting an indirect dual-task manipulation. As such, adopting a direct manipulation is likely to lead to a clearer focus on movement effects and increased efficiency of motor programming resulting in enhanced performance (Wulf & McNevin, 2003).

The vast majority of previous research has generally been conducted using movement tasks relevant to sport skills (e.g., golf, football, basketball), in which outcome was operationalized as the primary measure of performance (e.g., speed, accuracy, and movement amplitude). Consequently, it remains unclear whether the aforementioned findings generalize to other types of tasks, particularly those that can be classified as form tasks (i.e., gymnastics, karate kata, high diving). Furthermore, investigation of form-based tasks would help establish whether the enhanced outcome performance observed in the previous focus-of-attention literature is due, in part, to (a) the fact that previous research has used skills where movement effects have a direct and obvious impact on the environment (i.e., ball flight, golf club motion) or (b) are a result of more efficient movement technique.

First, if performance is to be enhanced, it is fundamental to both the action-effect principle (Prinz, 1997) and the CAH (Wulf et al., 2001) that attention must be directed toward a movement effect that is central to the goal of the action. The goal of the tasks adopted in the previous literature all result in the participants' movements having an obvious and direct impact on changing the environment (e.g., how the movement impacts on the trajectory of an object, be it a ball, balance platform, or golf club). As a result, the adopted external foci of attention are rather salient. However, it is unclear at present whether adopting an external focus will result in improved performance in tasks where the performers' actions result in little or no change to the environment (i.e., floor routines in gymnastics, movements in a karate kata).

Second, technique has been defined as "the motion activity specified by biomechanical principles of human motion which use motor features of movement and body structure to obtain the best performance" (Bober, 1981 p. 502). In addition, it has been suggested that it is implicit within the concept of technique that performance will be better if a skill is performed with "good" technique rather than "poor" technique (Lees, 2002). Thus, it is possible that the improved performance associated with adopting an external focus of attention is a direct result of enhanced technique. However, there is only a limited amount of research investigating the effects of attentional focus on form tasks and movement technique, with those that have examined technique doing so only as a secondary dependent variable (Wulf, McConnel, Gärtner & Schwarz, 2002). Here, participants performed volleyball serves and received feedback about their performance (on serve accuracy and movement technique) from either an internal or external focus. In line with previous research, the results revealed that the accuracy of the serve (i.e., the outcome performance measure) was enhanced in the external focus feedback group compared to that of the internal focus feedback group. However, contrary to these results, the technique and form of the participants' movements did not significantly differ between the two groups.

The goal of the current investigation was to extend our knowledge of an optimal focus of attention for novice performance in a form-based gymnastic task, the performance of which was measured on the production of movement techniques. To achieve this, participants learned a basic gymnastics routine under an external focus, an internal relevant focus, an internal irrelevant focus, or a control (no focus) condition. Consistent with Wulf and colleagues, we manipulated attentional focus through the use of explicit verbal instructions. However, because previous research has criticized the absence of manipulation checks within these designs (Castaneda & Gray, 2007), we also used a postmanipulation-check protocol (cf. Bell & Hardy, 2009) to ensure that conclusions based on the differing foci of attention could be attributed to the successful manipulation of participants' focus. In line with the findings for the form dependent variable of Wulf et al. (2002) and the suggestion that form-based tasks offer no obvious movement effect in terms of how performers' actions result in changes to the environment, it was hypothesized that the external focus group would result in similar performance during acquisition and learning compared to the internal foci groups and the control group. However, in line with the CAH, we hypothesized that the internal relevant focus group would result in lower performance compared to the internal irrelevant focus group, because the former would constrain the actions pertinent to accurate movement form and technique.

Method

Participants

Participants (M age = 20.3 years, SD = 1.6) with no previous experience in gymnastics volunteered to participate in this experiment (N = 40; 24 men and 16 women). All were naïve to the research hypotheses being tested, and they gave their informed consent before taking part in the investigation. The experiment was conducted in accordance with the institution's ethical guidelines for research involving human participants.

Task and Apparatus

The experiment took place in a laboratory in which two standard multipurpose gymnastics mats (2 m x 1 m x 50 mm) were set out in front of two white screens. Marker tape on the mats was used to identify the start position, and movements were recorded on a Sony Digital Video Camera Recorder (DCR-DVD106) mounted onto a tripod located at a distance 3.5 m from the participant and at an angle of 45° (0° was taken as the center of the participant's navel). At the start of the experiment, participants were shown a short video 10 times¹ on a television monitor (Aiwa VX-G142) of an expert gymnast performing a floor

routine. Participants were instructed to practice the routine so that they could reproduce the form as accurately as possible. The routine consisted of five simple movement components, each of comparable difficulty as listed by the Fédération Internationale de Gymnastique code of points (2007). Specifically these involved a starting position, a lunge, an arabesque, a full turn, and a finish position, which were all held for 3 s.²

After watching the video, participants were required to perform a pretest that consisted of one block of five trials. Following the pretest, participants were randomly allocated to one of four attentional focus groups; external, internal relevant, internal irrelevant, and control (no focus). Focus instructions directed participants on what to attend to but not where to look. Participants were also asked a simple question following each trial in order to reinforce the appropriate focus. The question depended on the group to which participants were assigned. Focus instructions were designed to be consistent with judging criteria. Additionally, instructions were constructed according to Wulf's (2007b) suggestions, which are based on Gentile's (1987) taxonomy of motor skills. Instructions were devised through consultancy with a UK Level 2 gymnastics club coach.

Focus Groups

Participants in the external focus group were instructed to focus on their movement pathway as well as to exert an even pressure onto the support surface. After every trial, participants were asked, "What was your movement pathway for the previous performance?" and "Did you exert an even pressure onto the support surface?" Participants were asked to respond with either "straight" or "not straight" and "yes" or "no," respectively. In the internal relevant focus group, participants were instructed to focus on exerting an equal force on their feet, keeping their arms out straight, level with their shoulders. Two reinforcing focus questions were asked: "Were your arms level with your shoulders during the previous performance?" and "Did you exert an equal force on your feet?" Participants responded with either "yes" or "no." Participants in the internal irrelevant focus group were instructed to focus attention on their facial muscles and facial expressions while performing the routine. In order to reinforce focus instructions, participants were asked, "What was your facial expression during the previous performance?" following each trial. Participants were asked to respond with either "happy," "sad," or "indifferent." Participants in the control group were given no instructions about focus.

Procedure

There were four main phases in the experiment: pretest, acquisition, retention, and transfer.³ The acquisition phase took place over two consecutive days. Day 1 con-

sisted of a pretest followed by four blocks of five acquisition trials, and Day 2 consisted of another four blocks of five acquisition trials. There was a 45-s break between trials, during which participants were asked their respective focus reinforcement question. Participants had a 5-min rest between blocks. Retention took place 1 week following acquisition and consisted of one block of five trials, during which there were no attentional focus instructions. Immediately following retention, participants completed a transfer test, again with no attentional focus instructions. During transfer, participants were to complete one block of five trials using the opposite foot and arm movements to those used during acquisition. For example, whereas participants previously began the routine standing on their right foot, they now began on their left foot.

To assess the extent to which participants were able to adopt the appropriate focus of attention, participants completed a postmanipulation questionnaire. Participants were asked to state the intensity of their focus on internal and external foci, revealing the extent to which they were able to adopt the appropriate focus of attention and also to what extent they adopted an inappropriate focus of attention (i.e., a focus that was different from the one that the participants' experimental group was instructed to adopt; see Table 1). During acquisition, the questionnaire was administered at the end of each day. It was decided that administering the manipulation check immediately following retention might act as a reminder of the focus instructions before transfer. Thus, a single (combined) manipulation check was conducted at the end of Day 3, following both retention and transfer. Adherence to the focus instructions for each of the three conditions was assessed using three separate 4 x 3 (Group [internal relevant, internal irrelevant, external, control] x Time [Day 1, Day 2, Day 3] mixed-model analyses of variance with repeated measures on the second factor (see Table 1).

Dependent Measures and Analyses

All trials were video recorded for analysis. Performance was assessed by two experienced, independent, gymnastics judges—Judge 1: British Gymnastic Association (BGA) qualified with 20 years of experience; Judge 2: BGA qualified with 8 years of experience. Both were blind to the research hypotheses and focus conditions and were absent during testing. Participants were judged according to the Fédération Internationale de Gymnastique Code of Points (2007) for Artistic Gymnastics (WAG/MAG). Judges viewed the video recordings and awarded points for each trial according to the criteria on the Code of Points, with marks deducted for poor execution and errors (see Table 2). A maximum score of 10 points could be awarded for the whole routine (this was a composite score for all five movements). Focus instructions in the internal relevant and external groups were designed to

provide equivalent performance benefits on these criteria. For example, the instructions in the internal relevant focus group were designed to affect the execution faults item for single bent arms (keeping your arms out straight...) and the criteria under the “maintaining balance” subheading (exerting an even pressure on your feet). For the external focus group, instructions were designed to aid the single-deviation-from-straight-direction item (movement pathway) and similar to the internal relevant group the criteria under the “maintaining balance” subheading (exerting an even pressure on the support surface). Because these items carry equal weighting (faults and penalties) for the routine adopted, performance benefits from the instructions should have been equal (see Table 2). The mean of each block of five trials was calculated to obtain an overall block score.

To assess reliability of judging, mean interjudge reliability scores were calculated and analyzed across all trials for both acquisition and learning (retention and transfer). The results of this analysis revealed a significant correlation ($r = 0.912$, $p < .001$), suggesting that participants' performance was rated similarly by both judges for each trial. In addition, the intrajudge reliability, as calculated by intraclass correlation (ICC), was analyzed using a random selection of five double-marked trials. Here the judge was blind to the fact that they were marking the same selection of trials twice. ICC was significant for both Judge 1, $F(4, 4) = 4171.80$, $p < .001$, and Judge 2, $F(4, 4) = 105.67$, $p < .001$.

To ensure there were no significant differences between the performances of the groups before testing, the means of pretest performance data were submitted to a one way (group) ANOVA. A 4×8 (Group [internal relevant, internal irrelevant, external, control] $\times$ Block [Block 1, 2, 3, 4, 5, 6, 7, and 8 of acquisition]) mixed-model ANOVA with repeated measures on the second factor was performed on the mean acquisition data. To assess the retention and transfer performance data mean scores were submitted to separate 4×2 (Group [internal relevant, internal irrelevant, external, control] $\times$ Block [pretest and retention or transfer, respectively]) mixed-model ANOVAs with repeated measures on the second

factor. In addition, two similar 4×2 (Group [internal relevant, internal irrelevant, external, control] $\times$ Block [acquisition and retention or transfer, respectively]) mixed-model ANOVAs with repeated measures on the second factor were conducted to assess possible changes from the end of acquisition to retention and transfer (the last block of acquisition trials were used in this analysis). Significant between-subject effects were broken down using Tukey's HSD post hoc tests, while significant within-subject effects were broken down into their simple main effects.

Results

Manipulation Check

Descriptive data for the manipulation check are presented in Table 3. In order to maintain brevity, a summary of all three separate analyses of the responses to each Likert-type attentional-focus question is reported. All revealed only a significant main effect of group: internal relevant question, $F(2, 27) = 12.64$, $p < .001$, $\eta^2 = .48$; internal irrelevant question, $F(2, 27) = 32.94$, $p < .001$, $\eta^2 = .71$; external question, $F(2, 27) = 42.36$, $p < .001$, $\eta^2 = .76$. Breakdown of each revealed that all participants reported greater Likert scores on the question most closely related to their required focus. For example, participants in the internal relevant condition scored significantly higher on the internal relevant focus question than participants in the external and internal irrelevant groups. Since there were also no main effects or interactions regarding time ($p > .05$), this pattern of results continued throughout the retention and transfer tests, where the focus instructions were not presented. Finally, the results also revealed that participants in the control group adopted both internal relevant and external attentional foci. Control group scores on the internal relevant item did not significantly differ from those reported by the internal relevant group, and those on the external item did not differ from those reported by the external group.

Table 1. Manipulation check questionnaire to assess the extent to which participants were able to adopt the appropriate focus of attention

Focus of attention	No focus	Weak	Moderate	Strong	Very strong
How intense was your focus on exerting an even pressure on your feet, keeping your arms out straight, level with your shoulders?	1	2	3	4	5
How intense was the focus to your facial muscles and expressions?	1	2	3	4	5
How intense was the focus to your movement pathway and maintaining an even pressure on the support surface?	1	2	3	4	5

Performance

Pretest. Analysis of the pretest data revealed no significant differences between the performance of the groups,

Table 2. General faults and penalties from the Fédération Internationale de Gymnastique Code of Points for artistic gymnastics (2007)

Faults	Sm. 0.10	Med. 0.30	Lge. 0.50	V. lge. 0.80
Execution faults				
Bent arms or bent knees				
Leg or knee separations				
Legs crossed during elements with twist				
Insufficient height of elements				
Insufficient exactness of tuck, stretch or pike position				
Failure to maintain stretched body posture				
Insufficient split				
Body posture in dance elements				
Hesitation during jumps, press or swing to handstand				
Insufficient extension in preparation for landing				
Precision (each movement has a clear start and finish position; each movement phase has to demonstrate perfect control)				
Throughout the entire exercise:				
Relaxed or incorrect foot/body/trunk posture				
Insufficient flexibility				
Insufficient dynamics				
Insufficient amplitude of elements				
Landing faults (all elements including dismounts)				
Deviation from straight direction				
Legs apart on landing				
Movements to maintain balance:				
Extra arm swings				
Additional trunk movements to maintain balance				
Extra steps, slight hop			Max 0.70	
Very large step or jump (guideline: 1 m)			Max 0.70	
Body posture fault				
Deep squat				
Brushing apparatus with hands/arms, but not falling against the apparatus				
Support on mat/apparatus with one or two hands				0.80
Fall on mat to knees or hips				0.80
Fall on or against apparatus				0.80

Note. Sm. = small; Med. = medium; Lge. = large; V. lge. = very large.

$F(3, 36) = 0.43, p = .73, \eta^2 = .36$ (external = 7.65, $SD = 0.37$; internal relevant = 7.51, $SD = 0.38$; internal irrelevant = 7.60, $SD = 0.34$; control = 7.60, $SD = 0.31$). Thus, any performance effects during or following acquisition could not be attributable to differences before the experiment.

Acquisition. The results revealed a significant main effect for block, $F(7, 252) = 3.84, p < .05, \eta^2 = .56$, but not group, $F(3, 36) = 0.76, p = .53, \eta^2 = .06$. Overall, there was a decline in performance from Block 1 to Block 4 (i.e., Day 1) and an increase in performance from Block 5 to Block 8 (i.e., Day 2); see Figure 1. There was also a significant Group $\times$ Block interaction, $F(21, 252) = 1.64, p < .05, \eta^2 = .25$. Breakdown of this interaction revealed a significant decrease in performance from Block 1 to Block 4 only for the external focus group. There was an increase in performance from Block 2 to Block 8 only for the internal irrelevant focus group.

Pretest Versus Retention. The analysis revealed only a significant main effect for block, $F(1, 36) = 10.65, p < .05, \eta^2 = .23$. Specifically, performance improved from pretest to retention (external = 7.88, $SD = 0.32$; internal relevant = 7.60, $SD = 0.24$; internal irrelevant = 7.94, $SD = 0.28$; control = 7.77, $SD = 0.30$), demonstrating learning in all groups. There was no significant main effect for group, $F(3, 36) = 1.99, p = .22, \eta^2 = .14$, nor a significant Group $\times$ Block interaction, $F(3, 36) = 0.36, p = .56, \eta^2 = .03$.

Pretest Versus Transfer. The results of the analysis comparing the pretest and transfer data (external = 7.78, $SD = 0.27$; internal relevant = 7.53, $SD = 0.24$; internal irrelevant = 7.61, $SD = 0.58$; control = 7.80, $SD = 0.31$) revealed no significant main effects for block, $F(1, 36) = 1.09, p = .30, \eta^2 = .30$, or group, $F(3, 36) = .94, p = .43, \eta^2 = .07$, and no significant interactions for Group $\times$ Block, $F(3, 36) = .25, p = .86, \eta^2 = .02$.

Acquisition Versus Retention. A significant main effect for block, $F(1, 36) = 6.95, p < .05, \eta^2 = .16$, revealed that performance at retention ($M = 7.80, SD = .30$) was greater than during acquisition ($M = 7.67, SD = .43$). There was no significant main effect for group, $F(1, 36) = 2.37, p = .16, \eta^2 = .17$, and no significant interaction, $F(3, 36) = 1.59, p = .27, \eta^2 = .12$.

Acquisition Versus Transfer. Results of the analysis comparing acquisition and transfer data revealed no significant main effects for block, $F(1, 36) = .00, p = .98, \eta^2 = .01$, or group, $F(3, 36) = .130, p = .29, \eta^2 = .07$, nor interactions for Group $\times$ Block, $F(3, 36) = .19, p = .86, \eta^2 = .12$.

Discussion

The objective of this investigation was to further examine optimal focus of attention for novice performance and to examine whether the previous attentional focus findings could be extended to a form sport. Typically, adopting an external focus of attention enhances

movement performance in a variety of motor skills (e.g. basketball, golf, jumping, and leg cycling). However, the present investigation failed to replicate these findings using a novel gymnastic routine. Attention was manipulated toward movements made by the individuals (either relevant to the skill or irrelevant to the skill) or the movement effects (external). No significant differences were found between the learning of the groups, as participants in all conditions improved performance at retention

and maintained performance at transfer. Thus, in the present form-based task, adopting either an external, an internal relevant, or an internal irrelevant focus of attention neither benefited nor degraded learning compared to a control group where no explicit focus-of-attention instructions were provided.

Because there was no benefit from adopting an external focus of attention during acquisition, retention, and transfer, it appears that the findings of the current inves-

Table 3. Means and standard deviations for the manipulation check questions as a function of group, the focus orientation of the manipulation check question

	External question			Internal relevant question			Internal irrelevant question		
	Day 1	Day 2	Day 3	Day 1	Day 2	Day 3	Day 1	Day 2	Day 3
External									
<i>M</i>	4.00	3.80	4.00	2.70	2.70	2.30	2.10	1.80	1.80
<i>SD</i>	0.47	0.63	0.47	0.48	0.67	0.82	1.10	0.92	0.79
Internal relevant									
<i>M</i>	2.80	2.50	2.30	3.80	4.00	3.70	1.70	2.00	1.60
<i>SD</i>	0.92	1.08	0.97	0.79	0.47	0.67	0.82	0.67	0.52
Internal irrelevant									
<i>M</i>	1.90	2.00	1.80	2.50	2.70	2.60	3.90	3.80	3.80
<i>SD</i>	0.74	0.67	0.63	0.97	1.16	0.70	0.99	0.63	0.63
Control									
<i>M</i>	3.40	3.50	3.40	3.10	3.50	3.60	2.10	2.50	2.40
<i>SD</i>	0.70	0.97	0.84	0.74	0.97	0.84	0.74	1.08	0.97

Note. External question: how intense was the focus to your movement pathway and maintaining an even pressure on the support surface? Internal relevant question: how intense was your focus on exerting an even pressure on your feet, keeping your arms out straight, level with your shoulders? Internal irrelevant question: how intense was the focus to your facial muscles and expressions? and testing phase (Day 1; at the end of Block 4, Day 2 at the end of Block 8, Day 3 at the end of transfer); *M* = mean; *SD* = standard deviation.

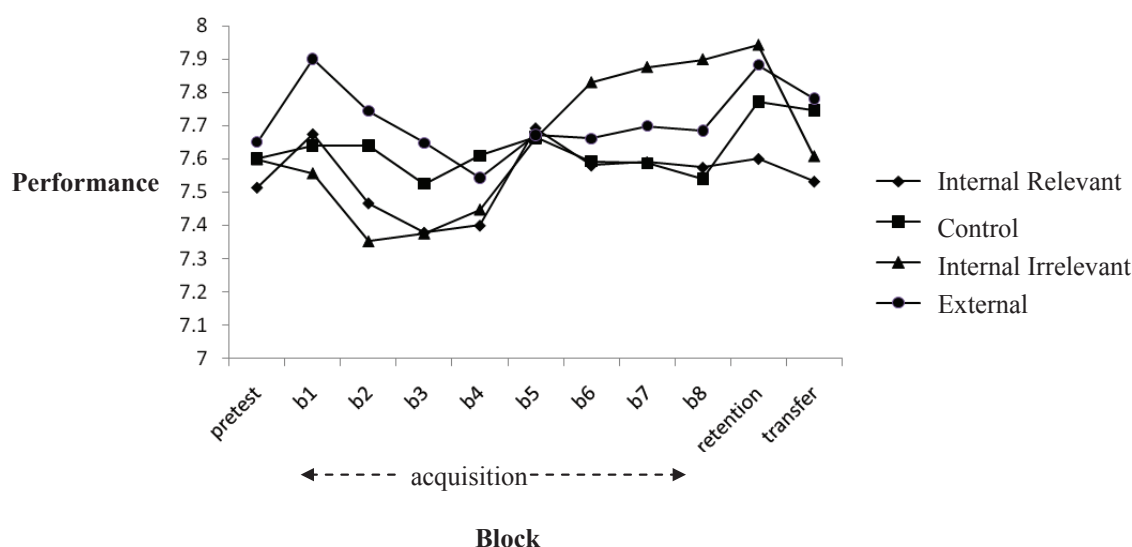


Figure 1. Mean performance (judges' scores) as a function of attentional focus and block (pretest, b1 = Trials 1–5; b2 = Trials 6–10; b3 = Trials 11–15; b4 Trials 16–20; b5 Trials 21–25; b6 = Trials 26–30; b7 = Trials 31–35; b8 = Trials 36–40, retention = Trials 41–45; transfer = Trials 46–50).

tigation do not support either the action-effect principle (Prinz, 1997) or the CAH (Wulf et al., 2001). While only the CAH can specifically account for the different learning effects of an internal versus an external focus of attention, both hypotheses agree that adopting an external focus of attention improves the efficiency of motor programming, through enhancing the congruence between movement planning and movement outcome. Specifically, Prinz's action-effect principle suggests that actions are planned in terms of their movement effects, with Wulf et al. (2001) rationalizing that when participants focus on movement effects the organization of motor programming is enhanced and performance is augmented. Thus, it is fundamental to direct attention toward a clear movement effect directly related to the goal of the action in order to enhance performance, regardless of which explanation one adopts. With this in mind, it is reasonable to suggest that if there is no obvious movement effect to attend to, then perhaps the benefit of adopting an external focus of attention is removed, a notion that the present findings appear to support.

The gymnastics floor routine used in this investigation was multifaceted and lacked an obvious movement effect in comparison to the tasks adopted in previous research. For example, the golf-putting and chipping tasks adopted in previous research (Bell & Hardy, 2009; Wulf et al., 1999; Wulf & Su, 2007) require only one discrete movement, with the desired outcome of placing the ball in or as close as possible to the target hole/marker. On the other hand, the gymnastics routine of the present investigation contained a number of linked, but distinctly different, movements (i.e., lunge to arabesque to full turn) with the desired outcome being to produce accurate form and look aesthetically pleasing (through the use of correct technique). As such, it is more challenging to direct attention to a clear movement effect that is both evident and suitable for the goal of each movement and the routine as a whole. Therefore, the tasks and corresponding verbal instructions used to manipulate an external focus of attention in previous research (i.e., "focus on the trajectory of the ball;" Bell & Hardy, 2009) may provide a more apparent focus toward the desired movement outcome when compared to the present investigation. Thus, focusing on the less apparent movement effects in the current task may have resulted in reducing the congruence between movement planning and movement outcome, and hence the benefits typically seen when adopting an external focus.

A null effect of focus of attention has been reported in previous research (Poolton, Maxwell, Masters, & Raab, 2006; Wulf, 2008). However, it is possible that participants did not adhere to the instructions requiring them to use a particular focus. The current investigation controlled for this by adopting a manipulation check based on that recently used by Bell and Hardy (2009). The results revealed that all participants adhered to the verbally pre-

sented focus instructions and consequently adopted the intended foci of attention during acquisition. It is interesting that, despite not being explicitly instructed to do so, participants reported adopting the same foci of attention during the retention and transfer phases of the experiment as to the foci required during acquisition. While it is beyond the scope of this investigation to determine whether these effects are due to either the presentation of the verbal focus instructions, the focus reminder questions, or a combination of the two, the findings do have some interesting implications from both a research and an applied perspective. First, Maxwell and Masters (2002) reported that participants switch attention according to specific task demands, thus highlighting the difficulty that researchers face when attempting to effectively manipulate attentional focus. Hence, from a research perspective, the present investigation's manipulation protocol appears to prevent or discourage this attentional switching and, importantly, enables participants to use and maintain the required experimental focus. Second, if coaches require performers to maintain a particular attentional focus adopted during practice in subsequent situations, perhaps where the delivery of focus instructions is not possible (i.e., competitions), then it is suggested that a focus manipulation similar to that of the present investigation be used during practice.

The current investigation revealed that performance improved from pretest to retention and hence learning occurred for all groups. However, there were no significant interactions involving group. Although there were no differences in learning between the different foci-of-attention manipulations as measured on retention, there was a significant difference in performance between the groups during acquisition. In line with the CAH (Wulf et al., 2002), it was hypothesized that the external focus group would improve performance to a greater extent than either the internal relevant, internal irrelevant, or control groups. However, results revealed that the external focus group's performance actually declined during the first half of acquisition and then leveled off, while only the internal irrelevant focus group enhanced performance (acquisition scores in this group increased from Trials 6–10 to Trials 36–40).

These findings are inconsistent with previous research that manipulated attention to external and internal features that differ in relation to their relevance to the skill (Ford et al., 2005). Ford et al. (2005) found that novice performers adopting an internal relevant focus performed better than when adopting an internal irrelevant focus. We propose that the conflicting results between the current experiment and those of Ford et al. (2005) are due to the increased task complexity and the resulting intricacy of the focus-of-attention manipulations in the present investigation. The task adopted by Ford et al. (2005) required participants to use a single foot to dribble a standard-size

football around a 9-m slalom course, with the primary dependent measure being the time taken to complete the course. In comparison, our gymnastics routine combined a series of five novel, linked but distinctly different, full body movements that were judged on whole body technique/form. As such, the ensuing focus instructions required the inclusion of a variety of items. For example, participants in the external focus group were asked to focus on their movement pathway and exert an even pressure onto the support surface, and those in the internal relevant group were asked to focus on exerting an equal force on their feet, keeping their arms out straight, level with their shoulders. It is possible that asking participants to focus on a number of features in conjunction with performing the complex attention-demanding gymnastics routine caused them to exceed their attentional capacity, and consequently a decline in performance occurred. In support of this suggestion, results of the manipulation check revealed participants in the control group adopted a combination of an internal and external focus and did not improve performance as typically seen in previous research (Beilock et al., 2002; Ford et al., 2005). Furthermore, the instructions of the internal irrelevant group asked participants to focus on a single item: their facial muscles. Thus, these instructions could be argued to be less attention-demanding compared to those adopted in the external, internal relevant, and (surprisingly) control groups, resulting in an improvement in performance only for the internal irrelevant focus group.

The results of this investigation do not support the findings from previous research involving outcome-based tasks, and the authors suggest that a possible reason is the reduced salience of movement effects when performing the current form-based task. However, the present findings are consistent with the only previous study that has explicitly assessed movement form (Wulf et al., 2002). In that study, the researchers examined the effect of directing attention through internal and external focus feedback on both the outcome and the form of a volleyball serve. In line with previous research, both novice and expert volleyball players demonstrated more accurate outcome scores when provided with the external focus feedback. Conversely, there were no significant differences between the two focus conditions in the participants' achievement of the correct movement form. When these results are combined with those of the present investigation, it appears that the benefits of adopting an external focus do not generalize to tasks that are assessed on the accurate production of a particular movement pattern/technique or form, where an obvious movement effect on the environment may be absent. However, it should be noted that in both experiments, form was measured using experienced judges and subjective judging criteria. This is not necessarily a limitation of the research per se, as this is often how the performance of form sports is measured in real sporting situations (e.g., gymnastics, high diving).

Nevertheless, we suggest that future studies consider using three-dimensional motion analysis techniques to give a more objective and detailed measure of form and technique when investigating the effects of adopting different foci of attention. For example, given the predictions of the CAH (Wulf et al., 2001), one would expect individuals performing a form task under an internal focus condition to constrain the body's degrees of freedom (see Turvey, 1977) to a greater extent than those in an external focus condition. Indeed this may have actually been the case in both Wulf et al. studies (2002a, 2002b) and in the current investigation; however, such patterns of results would likely be observed only through the use of more objective and detailed analysis techniques.

Conclusions drawn from the present investigation suggest that adopting either an internal or external focus of attention when performing a novel form sport does not result in any performance benefits. These findings may be due to both the increased complexity and ensuing intricacy of the focus instructions of the current task and the less salient movement effects associated with form actions. The former may result in an increase in the attention-demanding properties of two competing processes and produce a decline in performance, while the latter may ultimately reduce the congruence between movement planning and movement outcome. If adopting an external focus of attention only benefits actions with a clear movement effect on the environment, then it is imperative that coaches strive to determine these when directing both the learner's and potentially (although not explicitly investigated in the current experiment) the expert's attention, particularly in form-based tasks. It is also suggested that manipulation checks continue to be used in future investigations if concrete conclusions are to be drawn concerning the effectiveness of attentional foci. This is particularly important, as both external and internal foci have been shown to occur in control conditions.

References

- Al-Abood, S. A., Bennett, S. J., Hernandez, F. M., Ashford, D., & Davids, K. (2002). Effect of verbal instructions and image size on visual search strategies in basketball free throw shooting. *Journal of Sports Sciences*, 20, 271-278.
- Beilock, S. L., Carr, T. H., MacMahon, C., & Starkes, J. (2002). When paying attention becomes counterproductive: Impact of divided versus skill focused attention on novice and experienced performance of sensorimotor skills. *Journal of Experimental Psychology*, 8(1), 6-16.
- Bell, J. J., & Hardy, J. (2009). Effects of attentional focus on skilled performance in golf. *Journal of Applied Sports Psychology*, 21, 163-177.
- Bober, T. (1981). Biomechanical aspects of sports techniques. In A. Morecki, K. Fidelus, K. Kedzior, & A. Wit (Eds.), *Biomechanics VII* (pp. 501-509). Baltimore: University Park Press.

- Casteneda, B., & Gray, R. (2007). Effects of focus of attention on baseball batting performance in players of differing skill levels. *Journal of Sport and Exercise Psychology*, 29, 60–77.
- Deikman, A. J. (1966). De-automatization and the mystic experience. *Psychiatry*, 29, 324–338.
- Fasoli, S. E., Trombly, C. A., Tickle-Degnen, L., & Verfaellie, M. H. (2002). Effect of instructions on functional reach in persons with and without cerebrovascular accident. *American Journal of Occupational Therapy*, 56, 380–390.
- Fédération Internationale de Gymnastique. (2007). *Fédération Internationale de Gymnastique: Rules*. Retrieved from <http://www.sportcentric.com/vsite/vnavsite/page/directory/0,10853,5187-188050-205272-nav-list,00.html>
- Ford, P., Hodges, N. J., & Williams, A. M. (2005). Online attentional-focus manipulations in a soccer dribbling task: Implications for the proceduralization of motor skills. *Journal of Motor Behavior*, 37, 386–394.
- Gentile, A. M. (1987). Skill acquisition: Action, movement and neuromotor processes. In J. Carr & R. Shepherd (Eds.), *Movement science: Foundations for physical therapy in rehabilitation* (pp. 93–134). Rockville, MD: Aspen.
- James, W. (1890). *The principles of psychology* (Vol. 2). New York: Holt.
- Landers, M., Wulf, G., Wallmann, H., & Guadagnoli, M. A. (2005). An external focus of attention attenuates balance impairment in Parkinson's disease. *Physiotherapy*, 91, 152–185.
- Lees, A. (2002). Technique analysis in sports: A critical review. *Journal of Sports Sciences*, 20, 813–828.
- Maxwell, J. P., & Masters, R. S. W. (2002). External versus internal focus instructions: Is the learner paying attention? *International Journal of Applied Sports Sciences*, 14, 70–88.
- Perkins-Ceccato, N., Passmore, S. R., & Lee, T. D. (2003). Effects of focus of attention depend on golfer's skill. *Journal of Sports Sciences*, 21, 593–600.
- Prinz, W. (1997). Perception and action planning. *European Journal of Cognitive Psychology*, 9, 129–154.
- Poolton, J. M., Maxwell, J. P., Masters, R. S. W., & Raab, M. (2006). Benefits of an external focus of attention: Common coding or conscious processing? *Journal of Sports Sciences*, 24, 89–99.
- Totsika, V., & Wulf, G. (2003). An external focus of attention enhances transfer to novel situations and skills. *Research Quarterly for Exercise and Sport*, 74, 220–225.
- Turvey, M. T. (1977). Preliminaries to a theory of action with reference to vision. In R. Shaw & J. Bransford (Eds.), *Perceiving, acting and knowing* (pp. 211–265). Hillsdale, NJ: Erlbaum.
- Weeks, D. L., & Choi, J. (1992). Modeling the perceptual component of a coincident-timing skill: The influence of frequency of demonstration. *Journal of Human Movement Studies*, 23, 201–213.
- Wulf, G. (2007a). Attentional focus and motor learning: A review of 10 years of research. *Bewegung Und Training*, 1, 1–11.
- Wulf, G. (2007b). *Attention and motor skill learning*. Champaign IL: Human Kinetics.
- Wulf, G. (2008). Attentional focus effects in balance acrobats. *Research Quarterly for Exercise and Sport*, 79, 319–325.
- Wulf, G., Höß, M., & Prinz, W. (1998). Instructions for motor learning: Differential effects of internal versus external focus of attention. *Journal of Motor Behavior*, 30, 169–179.
- Wulf, G., Lauterbach, B., & Toole, T. (1999). Learning advantages of an external focus of attention in golf. *Research Quarterly for Exercise and Sport*, 70, 120–126.
- Wulf, G., McConnel, N., Gärtner, M., & Schwarz, A. (2002). Enhancing the learning of sport skills through external-focus feedback. *Journal of Motor Behavior*, 34, 171–182.
- Wulf, G., & McNevin, N. H. (2003). Simply distracting learners is not enough: More evidence for the learning benefits of an external focus of attention. *European Journal of Sport Science*, 3, 1–12.
- Wulf, G., McNevin, N. H., Fuchs, T., Ritter, F., & Toole, T. (2000). Attentional focus in complex motor skill learning. *Research Quarterly for Exercise and Sport*, 71, 229–239.
- Wulf, G., McNevin, N., & Shea, C. H. (2001). The automaticity of complex motor skill learning as a function of attentional focus. *The Quarterly Journal of Experimental Psychology*, 54, 1143–1154.
- Wulf, G., & Prinz, W. (2001). Directing attention to movement effects enhances learning: A review. *Psychonomic Bulletin and Review*, 8, 648–660.
- Wulf, G., & Su, J. (2007). An external focus of attention enhances golf shot accuracy in beginners and experts. *Research Quarterly for Exercise and Sport*, 78, 384–389.
- Wulf, G., Zachry, T., Granados, C., & Dufek, J. S. (2007). Increases in jump-and-reach height through an external focus of attention. *International Journal of Sports Science and Coaching*, 2, 275–284.
- Zachry, T., Wulf, G., Mercer, J., & Bezodis, N. (2005). Increased movement accuracy and reduced EMG activity as the result of adopting an external focus of attention. *Brain Research Bulletin*, 67, 304–309.

Notes

1. This frequency of demonstration was in accordance with Weeks and Choi (1992), who suggested this number was sufficient for the performer to create an accurate mental depiction of the skill.
2. The gymnastics routine consisted of five movements. For the start position, participants were required to balance on their right foot with their left leg bent and left foot resting on their right knee. They had to hold their arms horizontally in front of their body with their left arm at 45° and right arm straight out, with their hands and palms facing down and fingers straight. For the lunge, participants were required to step forward on their left foot and hold their right leg back straight with their body, arms horizontal in front of the body, and palms facing down. For the arabesque, participants were required to stand on their left leg, with their right leg behind, horizontal and straight, and foot pointed. They then had to circle their right arm back until it was straight behind the body, while holding the left arm horizontal and straight in front of the body, before returning to standing position. For the full turn, participants were required to jump in the air swinging their arms forward and overhead for momentum. Participants had to turn their head in the direction of

rotation (right), pulling with the opposite shoulder and hips to execute a 360° turn in the air, before landing on two feet, with their arms horizontal in front and palms down. The finish and starting positions were identical.

3. Wulf (2007a) suggested that retention tests do not always reflect true learning, as participants are likely to use the same instructions given during acquisition; thus, a transfer test was used in addition to the retention test to ensure that any learning effects were revealed.

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